

Uncertainty-Aware and Ordinally Calibrated Scoring for AI-Based Clinical Trial Matching

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Abstract

AI-based clinical trial matching systems increasingly support eligibility screening at scale, yet most approaches formulate eligibility as a strict binary decision [1, 2, 3]. In practice, patient-reported medical information is frequently incomplete or uncertain, causing binary screening systems to discard clinically relevant ambiguity and prematurely exclude potentially eligible patients [4, 5].

We propose a lightweight, uncertainty-aware eligibility scoring framework that represents patient responses using graded answers (yes / no / unsure) augmented with self-reported confidence levels. Responses are mapped to continuous eligibility scores via deterministic scoring functions, including a non-linear modulation of uncertainty that penalizes low-confidence responses while preserving informed uncertainty [6, 7]. The proposed design is empirically motivated and prioritizes interpretability and compatibility with existing trial matching pipelines. Our contribution is methodological rather than architectural. Holding all other components of the matching pipeline constant, we conduct a controlled ablation study comparing binary, graded linear, and graded quadratic scoring. Using a small manually constructed dataset for qualitative inspection and a large synthetic dataset ($N = 1000$) for statistical evaluation, we assess classification performance, ordinal behavior, and calibration properties [8, 9, 10].

Results show that graded scoring improves precision-recall trade-offs relative to binary baselines and produces eligibility scores with improved ordinal structure. In particular, graded quadratic scoring reduces Expected Calibration Error (ECE) by 44% (from 0.193 to 0.108) and improves Brier score (from 0.204 to 0.155) prior to post-hoc calibration. Bootstrap confidence intervals indicate that these calibration improvements are robust under resampling. These findings suggest that explicitly representing patient-reported uncertainty may improve eligibility scoring under controlled experimental conditions, pending validation in real-world clinical settings [11].

1 Introduction

Clinical trial recruitment remains a persistent bottleneck in drug development, contributing to delayed timelines, increased costs, and frequent trial termination due to insufficient enrollment [11]. Despite advances in automated screening and AI-assisted trial matching, eligibility assessment continues to rely heavily on manual interpretation of complex inclusion and exclusion criteria [12].

Recent work leveraging natural language processing (NLP) and large language models has improved the extraction and representation of eligibility criteria and patient information [13, 14, 15]. However, most systems ultimately reduce eligibility assessment to a binary decision [2, 3, 16]. This simplification introduces a fundamental methodological limitation: in real-world screening contexts, patient-reported information is often incomplete, uncertain, or imprecise, particularly for criteria involving technical diagnoses, laboratory thresholds, or historical medical events [5, 17].

Treating uncertainty as negative evidence risks systematically excluding potentially eligible patients and obscures clinically meaningful ambiguity [18]. In this work, we argue that eligibility scoring—often treated as a downstream heuristic—should instead be considered a first-class methodological design choice [19].

We propose a graded, uncertainty-aware eligibility scoring framework that explicitly captures patient uncertainty and produces continuous eligibility scores suitable for coarse prioritization and triage support. The goal is not to automate enrollment decisions, but to support clinical review through more informative and interpretable eligibility signals [4, 20].

2 Related Work

2.1 Clinical Trial Matching Systems

Early clinical trial matching systems relied on rule-based representations of eligibility criteria, enabling deterministic screening but exhibiting high sensitivity to missing or ambiguous information [12]. Subsequent machine learning approaches have improved matching accuracy through various architectural innovations [21], while natural language processing techniques have enhanced the extraction and understanding of eligibility criteria [22]. More recently, large language models have demonstrated promising capabilities for resource-efficient trial matching [23]. However, these advances largely retain binary eligibility formulations or rely on heuristic confidence scores with limited interpretability [1, 2, 24].

In contrast, our work focuses on the downstream scoring logic applied once eligibility questions and patient responses are defined, a component that remains underexplored despite its central role in decision support [15].

2.2 Uncertainty and Abstention

Uncertainty modeling and abstention have been studied extensively in machine learning [25, 26], including selective classification [27] and conformal prediction [28, 29]. These approaches typically provide formal coverage guarantees under specific assumptions (e.g., i.i.d. data).

In contrast, our work addresses epistemic uncertainty arising from patient self-reporting, where such assumptions may not hold [17]. We therefore prioritize a deterministic and interpretable scoring formulation that can be integrated into existing pipelines, viewing formal uncertainty guarantees as complementary directions for future work rather than direct baselines [4].

2.3 Calibration and Ordinal Validity

Calibration and ordinal coherence are critical when scores are used for triage or prioritization [10]. Early work established foundations for transforming classifier scores into accurate probability estimates [28], with subsequent research emphasizing the importance of well-calibrated probabilities in supervised learning [29]. While calibration has been extensively studied in clinical risk prediction [8, 9, 10], eligibility scores in clinical trial matching are rarely evaluated for calibration or monotonic behavior. Modern approaches have explored advanced calibration techniques beyond simple temperature scaling [30], yet these remain underutilized in trial matching systems. Our work explicitly addresses this gap.

3 Problem Formulation

We reformulate eligibility assessment as a continuous scoring problem rather than a binary classification task. Each eligibility question is answered using:

- an answer type: {yes, no, unsure}, and
- a self-reported confidence level on an ordinal scale from 1 to 5.

This representation allows patients to explicitly express epistemic uncertainty while indicating the strength of their belief [5]. Confidence is treated as an ordinal signal rather than a calibrated probability [6].

4 Uncertainty-Aware Eligibility Scoring

4.1 Question-Level Scoring

Each graded response is mapped to a score in the interval $[0, 1]$. Affirmative responses contribute proportionally to

confidence, negative responses contribute zero, and uncertain responses are modulated using a power-law transformation:

$$s(\text{unsure}, c) = 0.5 \times \left(\frac{c}{5}\right)^\alpha, \quad \text{with } \alpha = 1.5. \quad (1)$$

The exponent α was selected empirically based on a sensitivity analysis balancing penalization of low-confidence uncertainty and score stability. We do not claim a theoretical optimum and treat α as a tunable design parameter rather than a universal constant [6, 7].

4.2 Trial-Level Aggregation

Inclusion criteria are aggregated using the mean score across questions, while exclusion criteria are handled conservatively using a maximum rule. This reflects the clinical principle that a single strong exclusion criterion may suffice to disqualify a patient [19].

All scoring functions are deterministic and reproducible.

5 Experimental Design

We conduct a controlled ablation study comparing four scoring variants:

- Binary scoring
- Graded linear scoring
- Graded quadratic scoring (proposed)
- Binary scoring with post-hoc temperature scaling [8]

All experiments use identical patient cases, eligibility questions, aggregation logic, and thresholds, which were fixed a priori and not tuned on evaluation outcomes [10].

5.1 Datasets

Evaluation is performed on:

- a manual dataset of 15 carefully constructed cases for qualitative inspection of edge cases, and
- a synthetic dataset of 1000 cases for statistical evaluation.

Synthetic cases are generated using a rule-based scenario generator controlling the proportion of eligible, ineligible, borderline, and high-uncertainty cases. Ground-truth eligibility is assigned using predefined logical consistency rules rather than model predictions, ensuring that scoring functions are not evaluated against themselves. The synthetic dataset is intended as a methodological stress test for scoring behavior under controlled uncertainty scenarios, rather than as a realistic simulation of epidemiological prevalence [11].

6 Results

6.1 Classification and Calibration Performance

Table 1 summarizes the main performance metrics on the synthetic dataset.

Table 1: Performance and Calibration Comparison

Method	Acc	Prec	Recall	F1	ECE↓	Brier↓
Binary	0.656	0.634	0.878	0.736	0.193	0.204
Graded Linear	0.706	0.681	0.868	0.764	0.100	0.152
Graded Quadratic	0.688	0.679	0.815	0.741	0.108	0.155
Binary + Temp. Scaling	0.656	0.634	0.878	0.736	0.090	0.166

Graded linear scoring achieves the highest F1-score by prioritizing inclusivity, whereas graded quadratic scoring adopts a more conservative treatment of uncertainty, yielding slightly lower recall but improved calibration. Relative to graded linear scoring, graded quadratic scoring trades a -2.3% decrease in F1-score for a -44% reduction in ECE, reflecting an intentional design choice aligned with clinical screening contexts where downstream review costs may favor conservative uncertainty handling [10].

6.2 Calibration Analysis

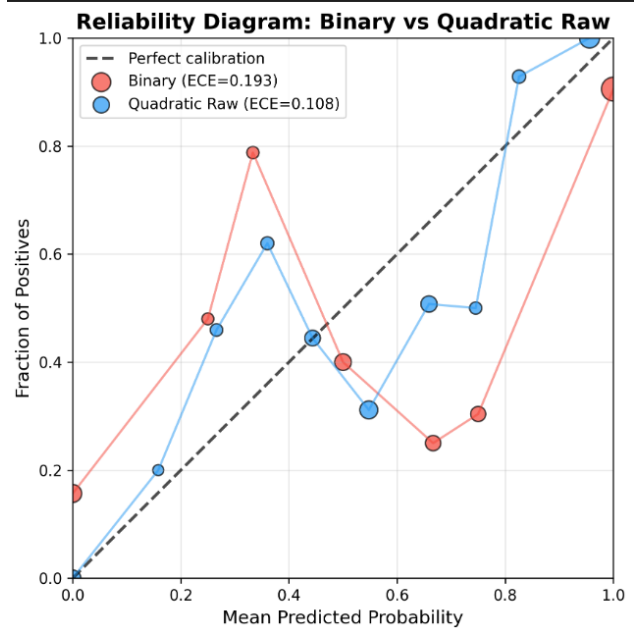


Figure 1: Reliability diagrams comparing Binary scoring (left) and Graded Quadratic scoring (right). The diagonal represents perfect calibration. Binary scoring shows systematic over-confidence in high-probability bins, while Graded Quadratic achieves better alignment between predicted and observed probabilities.

Binary scoring exhibits substantial miscalibration ($ECE = 0.193$, $Brier = 0.204$). Graded quadratic scoring achieves improved intrinsic calibration ($ECE = 0.108$, $Brier = 0.155$), corresponding to a 44% reduction in ECE relative to binary scoring prior to post-hoc calibration [8].

Temperature scaling reduces ECE for binary scores ($ECE = 0.090$) but yields a smaller improvement in Brier score (0.166 vs. 0.155). This suggests that post-hoc calibration and graded scoring address complementary aspects of calibration [30], with architectural choices in scoring functions providing intrinsic calibration benefits that post-hoc methods cannot fully replicate.

Bootstrap confidence intervals (95%) indicate that the calibration improvements of graded quadratic scoring over binary scoring are robust under resampling.

6.3 Ordinal Behavior

Score distribution analyses show that graded scoring produces smoother and more informative separation between eligible and ineligible cases, supporting its use for coarse prioritization and triage support rather than optimized ranking or hard decision-making [10].

7 Discussion

Our results indicate that explicitly representing patient-reported uncertainty can improve the interpretability and calibration of eligibility scores under controlled experimental conditions [4, 20]. Importantly, all observed improvements arise from scoring logic alone, without modifying retrieval models, NLP pipelines, or language model architectures [13, 14, 15].

Relative to graded linear scoring, graded quadratic scoring reflects a deliberate shift toward conservative uncertainty handling, trading a modest decrease in classification performance for improved calibration and reduced false-positive inflation [6].

We emphasize that practical deployment would require additional validation, including user studies assessing confidence elicitation and clinical workflow studies evaluating triage efficiency [5]. These are essential next steps beyond the scope of the present methodological evaluation [11].

8 Limitations

This study is methodological in nature and does not constitute clinical validation. The evaluation relies primarily on synthetic data, which cannot fully capture real-world clinical complexity [11]. The manually constructed cases serve only for qualitative inspection and are not intended to support quantitative claims.

We do not assess user experience, cognitive load, or clinical workflow integration [5]. In particular, we do not

claim that the reported calibration levels are sufficient for clinical deployment, but rather that graded scoring improves relative calibration properties under controlled experimental conditions [10].

Future work must validate the proposed approach on larger real-world datasets ($N \geq 100$) with documented annotation protocols and inter-rater agreement, and should complement distribution-level analyses with explicit ordinal metrics such as AUC or rank correlation to quantify prioritization behavior more formally [10].

9 Conclusion

We present a graded, uncertainty-aware eligibility scoring framework for AI-based clinical trial matching. Through controlled ablation and calibration analysis, we show that explicitly representing patient uncertainty improves ordinal behavior and calibration relative to binary baselines in controlled settings [8, 9, 10].

This work highlights eligibility scoring as a critical yet underexplored methodological component of clinical trial matching systems and provides a reproducible foundation for future validation in real-world clinical environments [11]. Future directions include integration with conformal prediction methods [29] and selective classification approaches [27] to provide formal coverage guarantees while preserving interpretability.

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Supplementary Material: Scoring Functions, Aggregation Rules, and Calibration Methods

Scoring Functions. Binary scoring assigns $s \in \{1, 0, 0.5\}$ for (yes, no, unsure). Linear and Quadratic ($\alpha = 1.5$) weight by confidence $c \in [1, 5]$: affirmative responses score $c/5$, uncertain responses score $0.5(c/5)$ and $0.5(c/5)^{1.5}$ respectively.

Confidence c	Binary	Linear	Quadratic
1	0.500	0.100	0.089
2	0.500	0.200	0.253
3	0.500	0.300	0.465
4	0.500	0.400	0.716
5	0.500	0.500	1.000

Quadratic scoring penalizes low-confidence uncertainty ($c = 1$: 0.089 vs. 0.100) while rewarding high-confidence uncertainty ($c = 4$: 0.716 vs. 0.400).

Trial-Level Aggregation. Inclusion criteria use the mean: $S_{\text{incl}} = \frac{1}{n} \sum_{i=1}^n s_i$. Exclusion criteria use a conservative maximum: $S_{\text{excl}} = \max_{j=1}^m e_j$. The final eligibility score combines both: $S_{\text{final}} = S_{\text{incl}} \times (1 - S_{\text{excl}})$.

Calibration Metrics. Expected Calibration Error (ECE) with $K = 10$ bins: $\text{ECE} = \sum_{k=1}^K \frac{|B_k|}{N} |\text{acc}(B_k) - \text{conf}(B_k)|$. Brier score: $\text{Brier} = \frac{1}{N} \sum_{i=1}^N (\hat{p}_i - y_i)^2$.

Post-Hoc Calibration. Temperature scaling [8] transforms probabilities as $\hat{p}_{\text{scaled}} = \sigma(\text{logit}(\hat{p})/T)$ with optimized temperature $T^* = 7.13$. Isotonic regression [28] fits a non-decreasing function achieving $\text{ECE} \approx 0$ and eliminating monotonicity violations.